

# HUM 347 — Complete Exam Preparation

## Full Model Answers (English) + Memory Guide

---

### QUESTION 1

**"Briefly discuss the ethical ideas of Socrates, Plato and Aristotle highlighting their views on morality, society and sense of good life. How are these classical ideas still relevant to modern technological society?"**

---

#### **PART A: The Classical Ethical Ideas**

**Introduction:** The Classical Period of ethical thought extends from 500 BC to 500 AD. During this time, Greek society was transforming from an agrarian monarchy into a commercial democracy.

Ethics was regarded as the "spiritual counterpart of medicine" — just as medicine cares for the body, ethics cares for the soul. Three dominant thinkers shaped this era: Socrates, Plato, and Aristotle.

---

#### **1. SOCRATES (469–399 BC) — Father of Western Philosophy**

Socrates is considered the dominant figure of the Classical Period and the father of Western philosophy. He left no writings; his ideas are known through his student Plato's works.

**On Morality:** Socrates rejected the Sophist idea that morality is merely tradition or personal convenience. He believed it is possible to develop a universal set of ethical principles through human reason. Morality, for him, was grounded in careful examination of beliefs and logical reasoning — not in custom or social agreement.

**On Society:** Socrates engaged society through his famous Socratic Method — rather than teaching directly, he asked fundamental questions such as "What is justice? What is virtue?" He exposed vagueness and inconsistency in people's answers, guiding them toward deeper inquiry. He was also the first to probe the relationship between facts and values.

**On the Good Life:** His most famous teaching was: "*The unexamined life is not worth living.*" The good life, for Socrates, meant continuous self-examination and self-knowledge ("Know thyself"). He believed in moderation — nothing should be done to excess.

---

## 2. PLATO (427–347 BC) — Student of Socrates

Plato built upon Socrates' teachings. His greatest work is *The Republic*, written in dialogue form with Socrates as the main character.

**On Morality:** Plato believed that morality is rooted in a higher reality — the world of abstract "Forms" or "Ideas." The concrete world we perceive through our senses is merely an imperfect reflection of this ideal world. The highest and most important Form is "**The Good**" — the ultimate moral reality. Thus, the goal of ethical life is to gain a vision of The Good.

**On Society:** In *The Republic*, Plato argued that an ideal society should be governed by a **Philosopher-King** — a wise ruler who has attained knowledge of The Good. Unlike Socrates who believed in moderation at all levels, Plato held a more extreme view — he completely denied bodily pleasure and sensory experience as paths to truth.

**On the Good Life:** Plato argued that intellectual pursuits are far superior to bodily pleasures. The good life means pursuing reason and wisdom, not physical satisfaction. This made him a model for later religious and idealistic ethicists.

---

## 3. ARISTOTLE (384–322 BC) — Student of Plato

Aristotle authored nearly 400 works on diverse subjects. His *Nicomachean Ethics* is the **first systematic treatment of ethics in Western civilization**.

**On Morality:** Aristotle rejected Plato's Theory of Forms, stating: "*No form without matter, no matter without form.*" He argued that moral principles are not found in a separate ideal world — they exist within the daily activities of human life and can be discovered by examining those activities.

**On Society:** Aristotle believed humans are naturally social beings — "*Man is a political animal.*" A good society creates conditions for individuals to develop virtue and achieve happiness. Unlike Plato's idealism, Aristotle's ethics was practical and grounded in real life.

**On the Good Life (Virtue Ethics):** Happiness (*eudaimonia*) is achieved by developing one's potential through a life of reason. Virtue, for Aristotle, represented a **midpoint (the "Mean") between two extremes**:

- Courage = midpoint between cowardice and recklessness
- Generosity = midpoint between miserliness and extravagance

The good life has two aims: the pursuit of truth through reflection, and the pursuit of virtue through intelligent conduct.

---

## PART B: Relevance to Modern Technological Society

The classical ideas of Socrates, Plato, and Aristotle remain profoundly relevant to modern technological society, particularly in the field of Computer Science and Engineering (CSE).

**Socrates → Relevant to Critical Thinking in Technology:** Socrates' method of questioning assumptions is the foundation of today's ethical technology auditing. Engineers must ask: "Does this software harm users? Is this algorithm fair?" Just as Socrates questioned society's moral assumptions, modern CSE professionals must question whether their systems uphold justice and virtue.

- *Example:* Facebook's algorithm promoting harmful content to teenagers — a Socratic examination would have identified this harm before deployment.
- The idea that "*the unexamined code is not worth deploying*" is the modern equivalent of Socrates' teaching.

**Plato → Relevant to Technology Leadership and Governance:** Plato's concept of the Philosopher-King translates into the need for **ethical technology leaders** — people who govern AI and digital platforms with wisdom and concern for the public good ("The Good"), not merely profit.

- *Example:* The Cambridge Analytica scandal demonstrated what happens when digital power is wielded without knowledge of The Good — 50 million Facebook users' data was misused for political manipulation.
- Plato's idea of pursuing The Good aligns with today's concept of **public interest technology**.

**Aristotle → Relevant to Balanced and Practical Ethics in CSE:** Aristotle's concept of the "Mean" is directly applicable to modern technological decisions. Extremes in technology are dangerous — both over-automation and zero automation cause harm. Ethical technology seeks the balanced middle path.

- *Example:* AI in hiring — complete automation (extreme) creates algorithmic bias; complete human discretion (other extreme) introduces human bias. A balanced, human-in-the-loop AI system reflects Aristotle's Mean.
- Aristotle's grounding of ethics in everyday life corresponds to the need for **practical, daily ethical coding practices** rather than abstract policy documents.

**Conclusion:** The classical thinkers laid the philosophical groundwork that modern technology ethics continues to build upon. Socrates taught us to question, Plato taught us to pursue the highest good, and Aristotle taught us to practice virtue in everyday professional life. For CSE professionals, these are not ancient relics — they are living guides to building technology that serves humanity.

---

## QUESTION 2

**"How does ethical awareness influence decision making in areas such as software development, data handling, artificial intelligence, or digital platforms? Use relevant examples."**

---

**Introduction:** Ethics is defined as the systematic study of what is morally right and wrong, good and bad, fair and unfair. In Computer Science and Engineering, ethical awareness means being

conscious of the moral dimensions of one's technical work. As CSE now shapes education, healthcare, finance, governance, and even warfare, ethical awareness is no longer optional — it is core to the identity of a responsible engineer.

Ethical awareness influences decision-making in CSE across four major areas:

---

### 1. Software Development:

Ethical awareness fundamentally changes how software is designed and tested. Without it, engineers focus only on functionality. With it, they ask: "Who might be harmed? Who is excluded? What are the unintended consequences?"

Key ethical decisions in software development include:

- Ensuring **accessibility** for people with disabilities
- Avoiding **dark patterns** (design tricks that manipulate users)
- Ensuring software does not discriminate based on gender, race, or religion

*Example:* Facebook's recommendation algorithm was found to promote content that caused depression and anxiety in teenage girls. Ethical awareness during development would have triggered safety testing for vulnerable users before deployment, preventing widespread psychological harm.

---

### 2. Data Handling:

Data is the foundation of modern technology. Ethical awareness guides engineers to ask: "What data should be collected? How long should it be stored? Who owns it? Who has access?"

Unethical data handling leads to **data breaches, surveillance, and manipulation**. Ethical data handling requires:

- **Informed consent** — users must know what data is collected
- **Data minimization** — collect only what is necessary
- **Secure storage** — protect data from unauthorized access

*Example:* The Cambridge Analytica scandal (2018) — 50 million Facebook users' personal data was harvested without consent and used to target political advertisements. This directly influenced the US presidential election. Ethical awareness would have prevented this misuse. Similarly, in Bangladesh, over 5 crore citizens' data was exposed online due to poor security practices — a direct consequence of inadequate ethical consideration in data handling.

---

### 3. Artificial Intelligence:

AI presents the most complex ethical challenges in modern CSE. Ethical awareness influences every stage — from dataset selection to deployment.

Major ethical issues in AI:

- **Bias in training data:** If the data used to train an AI model is biased, the AI will make biased decisions (systematic errors, imbalanced representation)

- **Discrimination in automated decisions:** AI making decisions based on gender, race, ethnic identity, or political belief
- **Black Box problem:** AI decisions that cannot be explained or challenged — lack of transparency
- **Job displacement:** Automation eliminating employment without ethical planning
- **Deepfakes and misinformation:** AI-generated content used to deceive

*Example:* Amazon's AI recruitment tool (2018) was found to systematically downgrade résumés from women because it was trained on historical hiring data that reflected past male dominance in tech. Ethical awareness at the dataset stage — checking for balanced, representative data — would have caught this bias before deployment.

*Example:* Facial recognition software showed racial bias — higher error rates for darker-skinned faces. AI experts challenged Amazon's facial recognition after a researcher found it misidentified Black individuals at significantly higher rates.

#### 4. Digital Platforms:

Social media and digital platforms have enormous power to shape public opinion, mental health, and political discourse. Ethical awareness influences how platforms are designed and regulated.

Key ethical issues:

- **Spread of harmful content** — misinformation, hate speech, violence
- **Manipulation through algorithms** — keeping users engaged through outrage and fear
- **Echo chambers and polarization** — algorithms showing only content that confirms existing beliefs, deepening societal division
- **Data harvesting for targeted advertising** — exploiting personal data without meaningful consent

*Example:* YouTube's recommendation algorithm was found to progressively guide viewers toward increasingly extreme content, contributing to radicalization. Ethical awareness would demand that recommendation systems be designed to prioritize user wellbeing over watch-time metrics.

**Conclusion:** Ethical awareness transforms CSE professionals from mere technical executors into responsible architects of society. Without it, technology amplifies inequality, invades privacy, and manipulates behavior. With it, technology can become a force for justice, inclusion, and human flourishing. As the slides state: "*Ethics is not optional — it is core to the identity of a responsible engineer.*"

# QUESTION 3

## "Compare Kant's Deontological Ethics and Mill's Utilitarianism. How are these Modern Period ideologies relevant to CSE?"

---

**Introduction:** The Modern Period of ethical thought (1500–present) marked a decisive shift from theological to scientific reasoning. Among the most influential modern ethicists are Immanuel Kant and John Stuart Mill, whose contrasting theories — **Deontology and Utilitarianism** — continue to shape professional ethics, including in CSE.

---

### PART A: Kant's Deontological Ethics (Ethics of Duty)

Immanuel Kant (1724–1804) is considered the basis of modern ethical discussion. His theory is known as **Deontology** — from the Greek *deon* meaning duty.

#### Core Principles:

- Morality must be grounded in **pure reason**, not emotion or consequences
- **"Good Will"** is the only thing that is unconditionally good — acting from a sense of duty is what gives an action moral worth
- Happiness is achieved through reason in a civilized society where humans are rational beings
- Natural law operates as **Categorical Imperative** — an absolute, universal moral command

#### The Three Formulations of the Categorical Imperative:

1. **Universal Law:** "Act only according to that maxim by which you can at the same time will that it should become a universal law." → Ask: If everyone did what I am about to do, would the world still function morally?
  2. **Humanity as End:** "Treat humanity, whether in your own person or in that of another, always as an end and never merely as a means." → Human beings have intrinsic worth — they must never be used as instruments for someone else's goals.
  3. **Duty Gives Moral Worth:** The moral value of an action comes from the intention and duty behind it — not the outcome it produces.
- 

### PART B: Mill's Utilitarianism (Ethics of Consequences)

John Stuart Mill (1806–1873) developed his ethics inspired by **James Mill and Jeremy Bentham**. His theory is called **Utilitarianism** — a form of **Teleology** (consequence-based ethics).

#### Core Principles:

- The rightness or wrongness of an action is determined entirely by its **consequences**
- The standard of moral judgment is **pleasure or happiness**

- The greatest ethical goal: "*The greatest happiness for the greatest number*"
- This is a **Hedonist ethics** — pleasure is the measure of morality

### Two Types of Motivation (Mill):

- **External motivation:** Acting because of friends' approval, social pressure, or fear of God's punishment
- **Internal motivation:** Acting from an inner feeling of humanity and conscience — learned from society

Mill argued that internal motivation — the deeply felt sense of duty to humanity — is the higher and more reliable moral guide.

### PART D: Relevance to CSE

#### Kant's Deontology in CSE:

Kantian ethics provides **absolute rules** that cannot be violated regardless of outcome — essential for establishing non-negotiable ethical boundaries in technology.

- *Universal Law:* If every engineer violated user privacy, the entire digital ecosystem would collapse. Therefore, privacy protection is a categorical duty.
- *Humanity as End:* Users must not be treated merely as data points or revenue sources. Designing systems that exploit psychological vulnerabilities (infinite scroll, notification manipulation) violates this principle.
- *Good Will:* Engineers should build secure systems because it is the right thing to do — not merely because regulations require it.

*Example:* A Kantian approach to data collection would prohibit selling user data under any circumstances — because using users as means to profit violates their dignity.

#### Mill's Utilitarianism in CSE:

Utilitarian ethics provides a **cost-benefit framework** for complex trade-offs in technology — useful when absolute rules produce impractical outcomes.

- *Greatest happiness:* A public health authority sharing anonymized patient data with researchers might violate individual privacy (harm to few) but could save thousands of lives (benefit to many) — a Utilitarian calculation might justify this.
- *Consequences matter:* An AI system that is 95% accurate but fails 5% of the time for a marginalized group causes measurable harm — the consequentialist calculus demands redesign.

*Example:* Open-source software releases security vulnerability information publicly to allow faster patching — some immediate risk to justify long-term greater security for all users. This is a Utilitarian decision.

**Which is More Applicable?** Neither alone is sufficient. Kantian ethics provides the **red lines** (user consent, data dignity, no discrimination), while Utilitarian ethics helps navigate **grey areas** (trade-offs between security and privacy, automation and employment). Mature CSE ethics requires both frameworks working together.

**Conclusion:** Kant and Mill represent two enduring pillars of moral thought that remain deeply relevant to modern CSE. Kant reminds engineers that some principles must never be compromised. Mill reminds them to always consider the real-world impact of their decisions on human happiness and welfare.

---

## QUESTION 4

**"Discuss how Social Ethics differs from Personal Morality. Why is this understanding important for CSE students?"**

---

**Introduction:** Ethics is the systematic study of what is morally right and wrong. Within ethics, an important distinction exists between personal morality — the individual's private moral code — and social ethics — the moral principles that govern collective behavior in society. For CSE students, understanding this distinction is foundational to becoming responsible professionals.

---

**Personal Morality:** Personal morality refers to an individual's own sense of right and wrong, shaped by personal experience, family upbringing, religion, and culture. It governs how a person behaves in private life.

Characteristics:

- Subjective — varies from person to person
  - Rooted in personal beliefs, religious values, and emotional responses
  - Applies to individual decisions and private conduct
  - Example: "I personally believe lying is wrong"
- 

**Social Ethics:** Social ethics refers to the moral principles and standards that govern behavior within communities, institutions, and professional settings. It is concerned with how individuals and organizations affect others and society at large.

Characteristics:

- Collective and institutional in nature
  - Rooted in social contracts, laws, cultural norms, and professional codes
  - Applies to decisions that affect groups of people
  - Example: "Organizations must not collect user data without informed consent"
- 

**Key Differences:**

| Aspect | Personal Morality          | Social Ethics                           |
|--------|----------------------------|-----------------------------------------|
| Scope  | Individual conduct         | Collective/institutional conduct        |
| Source | Personal belief, religion, | Social contract, law, professional code |



| Aspect         | Personal Morality                    | Social Ethics                                                |
|----------------|--------------------------------------|--------------------------------------------------------------|
|                | upbringing                           |                                                              |
| Accountability | To oneself or God                    | To society, institutions, law                                |
| Flexibility    | Personal and subjective              | More standardized and enforceable                            |
| Example in CSE | "I don't like writing insecure code" | "All software must meet security standards to protect users" |

---

### Why This Is Important for CSE Students:

A CSE student's future work will not remain private — it will affect millions of people. Personal morality alone is insufficient to navigate the complex ethical terrain of modern technology. Here is why:

- 1. Scale of Impact:** A single software application can reach millions of users globally. Personal morality governs one-to-one interactions; social ethics governs one-to-millions interactions.
- 2. Institutional Responsibility:** CSE professionals work within organizations, governments, and corporations. They must operate within ethical frameworks that go beyond personal preference — professional codes, regulations, and social accountability structures.
- 3. Conflict Between Personal and Social:** A CSE professional might personally disagree with a company's data practices but professionally be bound by social ethics to report or resist those practices (whistleblowing). Understanding the difference helps navigate such conflicts.
- 4. Technology Shapes Society:** Digital platforms, AI systems, and surveillance technologies reshape social norms and power structures. Personal good intentions are not enough — professionals must apply social ethical frameworks to ensure their work does not cause collective harm.

*Example:* An engineer may personally believe their facial recognition system is neutral and fair. But social ethics demands they test it against real-world demographic data to ensure it does not discriminate against racial minorities — because the social impact of deployment matters regardless of personal intent.

**Conclusion:** For CSE students, understanding the distinction between personal morality and social ethics is the first step toward professional ethical maturity. Technology is a social force — and its creators bear social responsibility.

---

## QUESTION 5

**"Compare the Classical, Medieval, and Modern Periods of ethical thought and their relevance to CSE."**

---

**Introduction:** Historians divide the history of Western ethics into three major periods: the Classical (500 BC–500 AD), the Medieval (500–1500), and the Modern (1500–present). Each period reflects

the dominant intellectual, social, and spiritual concerns of its time, and each contributes something essential to contemporary CSE ethics.

---

## THE CLASSICAL PERIOD (500 BC – 500 AD)

Context: Greek society was transitioning from agrarian monarchy to commercial democracy. Ethics was called the "spiritual counterpart of medicine" — caring for the soul as medicine cares for the body. The moral focus was the individual's duty as a citizen.

Key Thinkers and Ideas:

- **Socrates:** Morality grounded in reason; Socratic Method; "The unexamined life is not worth living"; self-knowledge as ethical foundation
  - **Plato:** Theory of Forms; "The Good" as highest reality; intellectual pursuits over bodily pleasure; Philosopher-King; *The Republic*
  - **Aristotle:** Virtue as the mean; ethics rooted in daily life; happiness through reason; *Nicomachean Ethics* — first systematic ethics text; "No form without matter"
  - **Other schools:** Hedonism (Cyrenaics, Epicureans) — pleasure as measure; Stoics and Cynics — duty and reason over desire
- 

## THE MEDIEVAL PERIOD (500 – 1500)

Context: Christianity dominated Europe. The citizen's relationship to the state was replaced by the individual's obligation to God. Medieval ethics combined classical reason with religious faith and obedience to God's will.

Key Thinkers and Ideas:

- **St. Augustine (354–430):** Combined Platonic philosophy with Christian ethics; two-sided system — reason leads to temporal good, faith leads to eternal salvation; original sin but retained free will; virtue requires faith to have real value; *Confessions*, *The City of God*
  - **St. Thomas Aquinas (1225–1274):** Rediscovered and Christianized Aristotle; ethics has two dimensions — natural (reason → earthly happiness) and theological (faith → eternal life); Natural Law theory — divine law written in the human heart, discoverable by reason; harmony of reason and faith; *Summa Theologica*; prepared the way for scientific ethics
- 

## THE MODERN PERIOD (1500 – Present)

Context: Protestant Reformation challenged Church authority; printing press democratized knowledge; scientific revolution (Copernicus, Galileo, Harvey) shifted focus from theology to science; individual reasoning became central.

Key Thinkers and Ideas:

- **Thomas Hobbes (1588–1679):** First to approach ethics scientifically; mechanistic materialism; human nature is self-interested; morality arises only through social contract; sovereign authority ensures order; *Leviathan*

- **Reaction to Hobbes:** Humans have moral faculty — intuition, moral sense, conscience — beyond mere self-interest
- **David Hume (1711–1766):** Moral judgment involves feeling; two-sided standard — objective consequences + subjective pleasure; moral sentiment over pure reason; naturally inclined to benefit others
- **Immanuel Kant (1724–1804):** Deontology — ethics of duty; Good Will; Categorical Imperative; morality grounded in pure reason; humans as rational beings
- **John Stuart Mill (1806–1873):** Utilitarianism — ethics of consequences; greatest happiness for greatest number; hedonist ethics; internal and external motivation

---

### Comparative Summary:

| Aspect             | Classical                  | Medieval            | Modern                    |
|--------------------|----------------------------|---------------------|---------------------------|
| Central Focus      | Citizen & city-state       | God & Church        | Individual & science      |
| Foundation         | Human reason               | Faith + reason      | Scientific reason         |
| Source of morality | Reason & virtue            | God's will          | Duty / consequences       |
| Key question       | What is the good life?     | What does God will? | What is the right action? |
| Key thinkers       | Socrates, Plato, Aristotle | Augustine, Aquinas  | Hobbes, Hume, Kant, Mill  |

---

### Relevance to CSE:

Classical → Questioning assumptions, virtue in everyday coding practice, balance (Aristotle's Mean) in technology deployment

Medieval → Natural Law foreshadows modern international cyber law and universal human rights in technology; the idea that ethics is accessible through reason supports secular technology ethics frameworks

Modern → Kant's Categorical Imperative informs data ethics (user consent, privacy as duty); Mill's Utilitarianism informs cost-benefit analysis in AI deployment and policy; Hobbes' Social Contract theory underpins digital governance and cybersecurity law

**Conclusion:** The three periods collectively form the philosophical heritage of modern CSE ethics. Classical thinkers gave us the questions; medieval thinkers embedded ethics in universal law; modern thinkers gave us the frameworks we still use to answer ethical dilemmas in technology today.

---



---

## MEMORIZATION GUIDE — How to Remember Everything

---

## STRATEGY 1: The "SPA" Framework for Classical Period

Remember the 3 classical philosophers as **SPA**:

S → Socrates → Self-examination → "Question everything"  
P → Plato → Perfect Forms → "The Good is real"  
A → Aristotle → Activity/Daily → "Virtue is the Mean"

For each philosopher remember **3 things: Person → Key idea → Famous phrase**

| Person    | Key Idea                | Famous Phrase/Concept |
|-----------|-------------------------|-----------------------|
| Socrates  | Reason + Self-knowledge | "Unexamined life..."  |
| Plato     | Forms + The Good        | "The Republic"        |
| Aristotle | Virtue as Mean          | "Nicomachean Ethics"  |

---

## STRATEGY 2: The "3 C's" for Historical Periods

Classical → CITIZEN (duty to the city-state)  
Christian → CHURCH (duty to God) = Medieval  
Current → CONSCIENCE (duty to reason/science) = Modern

---

## STRATEGY 3: The "K vs M" Shortcut for Modern Period

KANT = K = "Keep the rule" (Duty-based, never break rules)  
MILL = M = "Measure the result" (Consequence-based, count happiness)

To remember Kant's Categorical Imperative — "**U-H-D**":

- **U**niversal law — Would it work if everyone did it?
  - **H**umanity as end — Never use people as tools
  - **D**uty gives worth — Intention matters, not result
- 

## STRATEGY 4: Story Method for Medieval Thinkers

**Augustine's Story:**

Augustine was a wild young man who converted to Christianity. He combined his old love of Plato with his new faith → Plato + Christianity = Augustine. He said: "This world is temporary, heaven is real."

**Aquinas's Story:**

Aquinas arrived and said "Wait — people forgot Aristotle!" He brought Aristotle back and added Christianity → Aristotle + Christianity = Aquinas. He said: "Reason and faith are friends, not enemies."

Memory hook: "**A before Q**" = **A**ugustine came before **A**quinas (chronologically)

- Augustine = A = Earlier (354–430)

- Aquinas = Q (Thomas aQuinas) = Later (1225–1274)
- 

## STRATEGY 5: Acronym for CSE Ethical Issues

Remember "DC-SAI-WE" (think: "DC's AI world"):

D → Data Privacy & Surveillance  
C → Cybersecurity  
S → Social Media Ethics  
A → AI & Algorithmic Fairness  
I → Intellectual Property  
W → Workplace Ethics  
E → Environmental Sustainability

---

## STRATEGY 6: The "BURN" Method for Exam Day

Before writing any answer, quickly write on scratch paper:

B → Background (Which period? Which thinker? Context?)  
U → Understanding (What is the core idea/theory?)  
R → Relevance (How does it connect to CSE/modern world?)  
N → Now (Give a real-world example)

This ensures every answer has introduction, theory, application, and example — the full 15-mark structure.

---

## STRATEGY 7: Dates Anchor System

Anchor dates to memorable events:

| Thinker   | Dates      | Memory Trick                                           |
|-----------|------------|--------------------------------------------------------|
| Socrates  | 469–399 BC | "4-6-9 = almost like 1-2-3 backwards"                  |
| Plato     | 427–347 BC | "Plato's plate has 4 corners (427)"                    |
| Aristotle | 384–322 BC | "384 = 3 before 8 before 4"                            |
| Augustine | 354–430    | "Augustine came after Rome fell (430 AD = Rome falls)" |
| Aquinas   | 1225–1274  | "12 hundred" — Medieval middle                         |
| Hobbes    | 1588–1679  | "1588 = Spanish Armada year — easy anchor"             |
| Kant      | 1724–1804  | "Kant CAN'T be remembered — 1-7-2-4, just memorize"    |
| Mill      | 1806–1873  | "Mill ran in the 1800s"                                |

---

## STRATEGY 8: One-Line Summaries (Perfect for Quick Revision)

Write these on a small card and read before sleeping:

**Socrates:** "Question everything with reason — the unexamined life is not worth living."

**Plato:** "The highest reality is The Good — pursue it through intellect, not pleasure."

**Aristotle:** "Virtue is the mean — ethics lives in daily life, not abstract ideals."

**Augustine:** "Reason gives earthly good; faith gives eternal good — this life is only preparation."

**Aquinas:** "Reason and faith are in harmony — Natural Law is God's law written in our hearts."

**Hobbes:** "Humans are self-interested — only social contract and sovereign authority create morality."

**Kant:** "Act from duty — treat humans as ends, follow universal law, let Good Will guide you."

**Mill:** "Judge by consequences — the greatest happiness for the greatest number is the moral goal."

---

## QUESTION 6

**"What is Ethics? Discuss its key definitions, origin, and significance."**

---

### **PART A: The Concept and Definitions of Ethics**

**Origin of the Word:** The word "Ethics" derives from the Greek word '**Ethos**', which means "the characteristic spirit or attributes of a community, people, or system." From this root, ethics developed into a formal academic discipline concerned with understanding the moral dimensions of human life and conduct.

**Core Definition:** Ethics is the **systematic study of what is morally right and wrong, good and bad, fair and unfair**. It provides principles to guide human behavior, social interaction, and professional practice. In the academic journey — especially in sociology and engineering — ethics helps students understand how decisions impact individuals and society.

#### **Four Key Definitions of Ethics:**

**Definition 1 — Ethics as a set of moral principles:** Ethics is a collection of moral principles that guide human conduct. These principles distinguish acceptable behavior from unacceptable behavior in a given community or profession. They are not merely personal preferences but structured, reasoned standards.

**Definition 2 — Ethics as the art of human living:** Ethics is described as "the art of human living" — meaning it is not simply a theoretical subject but a practical guide for how to live well, make good decisions, and interact justly with others in everyday life.

**Definition 3 — Ethics as the formation of good character:** Ethics intends to form good men and women. The ultimate purpose of ethical study and practice is character formation — developing individuals who consistently choose what is right, not because they are forced to, but because they have internalized moral values.

**Definition 4 — Ethics as systematic analysis of morality:** Ethics is the scientific evolvment and demonstration of morality — meaning it applies rigorous, systematic analysis to moral questions rather than relying on intuition, tradition, or emotion alone.

**The End Goal of Ethics:** The final product of ethical development should be **ease in being a good person in every circumstance** — a person who does not struggle to choose right from wrong but acts morally as a matter of character and habit.

---

**What Ethics Studies:** Ethics provides principles to guide:

- Human behavior in personal life
- Social interaction in community
- Professional practice in careers

In the academic context, particularly in engineering and sociology, ethics helps students understand:

- How decisions impact individuals and society
  - What responsibilities professionals carry
  - Why moral reasoning matters in modern life
  - How social norms shape our sense of right and wrong
- 

**What Ethics Is NOT:** Ethics is not simply following laws — laws set minimum standards, but ethics demands higher reflection. Ethics is not merely cultural custom — customs vary, but ethical reasoning seeks universal principles. Ethics is not emotion alone — while feelings matter, ethics requires reason and analysis.

---

## **PART B: Significance of Ethics in CSE**

Understanding what ethics is becomes especially critical in Computer Science and Engineering, because CSE now shapes nearly every sector — education, communication, health, finance, public governance, surveillance, and even warfare.

**Why Ethics matters in CSE:**

Ethics is **not optional** — it is core to the identity of a responsible engineer. Ethical literacy helps future engineers:

- Understand the societal impact of their work
- Identify harms and unintended consequences before they occur
- Make decisions that protect people, organizations, and the environment
- Build public trust in technology

Without a clear understanding of what ethics is and why it matters, a CSE professional becomes a technically skilled but morally unguided force — capable of immense harm despite good intentions.

*Example:* The Cambridge Analytica scandal demonstrated what happens when engineers build powerful data tools without understanding ethics — 50 million Facebook users' data was harvested and used to manipulate elections. The engineers involved were technically skilled but ethically unaware.

*Example in Bangladesh:* Over 5 crore Bangladeshi citizens' personal data was exposed online due to poor security practices — a direct result of engineers not understanding their ethical responsibility to protect user data.

**Conclusion:** Ethics is the foundation of responsible human life and professional practice. For CSE students, understanding what ethics is — its origin, its definitions, and its goals — is the first step toward becoming engineers who build technology that serves rather than harms humanity. As the definition states: the end product of ethics should be ease in being a good person in every circumstance — including every professional circumstance an engineer faces.

---

## QUESTION 7

**"How can CSE professionals contribute ethically to social development in Bangladesh? Reflect on issues such as digital divide, data privacy, social media use, and automation."**

---

### **PART A: Ethical Issues Relevant to Bangladesh's Social Development**

Bangladesh is a rapidly digitalizing nation. While technology offers enormous opportunity — mobile banking, e-government, digital education — it also brings serious ethical challenges that CSE professionals are uniquely positioned to address. Understanding these issues is the foundation for ethical contribution.

**Issue 1 — Digital Divide:** The digital divide refers to the gap between those who have access to digital technology and those who do not — divided along lines of income, geography, gender, and education. In Bangladesh, this divide is particularly stark between urban centers like Dhaka and rural areas, and between men and women.

**Issue 2 — Data Privacy:** Bangladesh has experienced severe data privacy failures. In 2023, over 5 crore (50 million) Bangladeshi citizens' personal data — including NID information — was found exposed online due to inadequate security in government systems. This represents a massive ethical failure in data handling.

**Issue 3 — Social Media Use:** Social media platforms like Facebook are widely used in Bangladesh but have been implicated in the spread of misinformation, hate speech, and communal violence. Algorithmic amplification of outrage content has real-world consequences in the Bangladeshi social context.



**Issue 4 — Automation:** As automation increasingly enters Bangladesh's garment sector — the backbone of its economy — millions of low-skilled workers, predominantly women, face the threat of job displacement. The ethical implications of deploying automation without social safety nets are profound.

---

## **PART B: How CSE Professionals Can Contribute Ethically**

### **1. Bridging the Digital Divide:**

CSE professionals can contribute by designing technology that is **inclusive by default** — accessible to low-income users, available in Bangla, and functional on low-end devices with limited connectivity.

Ethical contributions include:

- Developing **low-bandwidth applications** for rural users
- Creating **Bangla-language digital content** and interfaces to reduce language barriers
- Building **affordable tech solutions** for small farmers, rural health workers, and local entrepreneurs
- Advocating for **universal internet access** as a social right

*Example:* bKash — a Bangladeshi mobile financial service — extended financial services to millions of unbanked citizens. CSE professionals who built bKash with accessibility in mind contributed directly to financial inclusion and social development.

### **2. Protecting Data Privacy:**

CSE professionals must treat data protection as an ethical duty, not merely a regulatory requirement. In Bangladesh, where data protection legislation is still developing, engineers must be proactive.

Ethical contributions include:

- Designing systems with **privacy by design** — building protection into architecture from the start, not adding it later
- Implementing **strong encryption** and **access controls** for systems handling citizens' data
- Advocating for **informed consent mechanisms** — ensuring citizens know what data is collected and how it is used
- Reporting data vulnerabilities through proper channels rather than ignoring them

*Example:* The 2023 Bangladeshi data breach could have been prevented if the engineers who built the government database had implemented basic security standards and conducted regular vulnerability testing. CSE professionals who understand their ethical responsibility would have raised alarms before public exposure.

### **3. Responsible Social Media Design and Use:**

CSE professionals working on platforms used in Bangladesh — or building local platforms — have an ethical responsibility to prevent their systems from becoming tools of division and violence.

Ethical contributions include:

- Designing **content moderation systems** sensitive to the Bangladeshi social and political context
- Building **fact-checking integrations** to slow the spread of misinformation
- Avoiding algorithmic designs that prioritize engagement over truth — echo chambers and polarization must be actively countered
- Collaborating with civil society organizations to understand how platforms are being misused

*Example:* During periods of communal tension in Bangladesh, Facebook has been used to spread false information that triggered real-world violence. CSE professionals who understand social media ethics could design algorithmic safeguards that reduce the viral spread of unverified content in sensitive periods.

#### 4. Ethical Automation:

As automation enters Bangladesh's industrial sectors, CSE professionals must balance technological efficiency with human welfare.

Ethical contributions include:

- Conducting **social impact assessments** before deploying automation in labor-intensive sectors
- Designing automation systems that **augment rather than replace** human workers where possible
- Collaborating with policymakers to design **reskilling and retraining programs** for displaced workers
- Refusing to deploy automation systems that disproportionately harm marginalized groups — particularly women in the garment sector — without adequate social protection

*Example:* A CSE professional designing an automated quality control system for a garment factory should simultaneously advocate for worker retraining programs so that displaced workers can move into operating or maintaining the new systems rather than simply losing their livelihoods.

---

**Conclusion:** CSE professionals in Bangladesh carry a unique responsibility at the intersection of technology and social development. Their technical skills give them the power to either deepen inequality or dismantle it, to either expose citizens to harm or protect them. Ethical contribution requires more than technical competence — it requires social awareness, moral courage, and a commitment to the public good. As ethics teaches us: the goal of moral life is ease in doing good in every circumstance — and for CSE professionals in Bangladesh, that means building technology that uplifts society rather than exploits it.

---

# QUESTION 8

## "What is Kant's Categorical Imperative? Explain its three formulations and discuss its relevance to CSE ethics."

---

### PART A: Kant's Ethical Philosophy and the Categorical Imperative

**Background — Immanuel Kant (1724–1804):** Immanuel Kant is considered the basis of modern ethical discussion. He belongs to the Modern Period of ethical thought (1500–present), during which individual reasoning and scientific thinking replaced theological authority as the foundation of ethics. Kant's ethical theory is called **Deontology** — from the Greek *deon* meaning duty.

#### Core Foundations of Kant's Ethics:

Kant argued that morality must be grounded in **pure reason**, not in emotion, tradition, or consequences. He believed that:

- **"Good Will"** is the only thing that is unconditionally good. An action has moral worth only when it is done out of duty — from a good will — not from inclination, desire, or self-interest.
- Happiness will be achieved through reason — not through the pursuit of pleasure or emotional satisfaction.
- In a civilized society, human beings are **rational beings** — and this rationality is the basis of their dignity and moral standing.
- Natural law operates as **categorical imperativeness** — absolute moral commands that apply universally, without exception.

**What is the Categorical Imperative?** The Categorical Imperative is Kant's supreme principle of morality — a universal moral law derived from reason that applies to all rational beings in all situations, without condition or exception. It is "categorical" because it commands unconditionally — unlike a "hypothetical imperative" which says "do X if you want Y," the Categorical Imperative says "do X — full stop."

---

#### The Three Formulations of the Categorical Imperative:

**Formulation 1 — The Universal Law:** *"Act only according to that maxim by which you can at the same time will that it should become a universal law."*

Before acting, ask: "What if everyone in the world did exactly what I am about to do?" If the world would collapse morally or practically, then the action is wrong. If it would function perfectly, then the action is permissible.

Example: If I am considering lying — ask: "What if everyone lied all the time?" Communication would break down, trust would disappear, and society would collapse. Therefore lying is categorically wrong, regardless of convenience or outcome.

**Formulation 2 — Humanity as End:** *"Act so that you treat humanity, whether in your own person or in that of another, always as an end and never as a means only."*

Human beings have intrinsic dignity and worth — they must never be treated merely as instruments or tools for achieving someone else's goals. Every person is an end in themselves.

Example: Using someone's emotional vulnerability to sell them a product they do not need treats them as a means to profit — this violates their dignity as a rational being.

**Formulation 3 — Duty Gives Moral Worth:** The moral worth of an action comes entirely from the **duty and intention** behind it — not from the result it produces. A person who tells the truth only because it is convenient acts from self-interest. A person who tells the truth because it is their duty acts morally, even if the outcome is painful.

---

## **PART B: Relevance of Kant's Categorical Imperative to CSE Ethics**

Kant's Categorical Imperative provides CSE with **absolute ethical boundaries** — non-negotiable rules that protect human dignity regardless of business pressure, technical capability, or expected outcomes.

### **Formulation 1 — Universal Law in CSE:**

Ask: "What if every CSE professional did what I am about to do?"

- If every engineer collected user data without consent — the entire digital ecosystem would collapse. No one would trust any platform. Therefore, unauthorized data collection is categorically wrong.
- If every developer introduced backdoors into security systems "just to see if it works" — cybersecurity would be universally compromised. Therefore, this is categorically impermissible.
- If every company sold user data to the highest bidder — users would be permanently exposed to manipulation. Therefore, selling personal data without explicit consent violates the Universal Law.

### **Formulation 2 — Humanity as End in CSE:**

Users are not data points. Citizens are not revenue streams. Communities are not test subjects.

- **Social media design:** Designing algorithms that exploit psychological vulnerabilities — fear, outrage, loneliness — to maximize engagement treats users as means to profit. This is a categorical violation of human dignity.
- **AI systems:** Using communities of color as test subjects for facial recognition without their knowledge or consent — as happened with several US police departments — treats those communities as means, not ends.
- **Targeted advertising:** Harvesting intimate personal data to manipulate purchasing decisions exploits human rationality rather than respecting it.

*Example:* Cambridge Analytica used psychological profiling to manipulate voters' political behavior without their knowledge — treating 50 million people as means to a political end. This is one of the clearest Kantian violations in the history of technology ethics.

### Formulation 3 — Duty in CSE:

Engineers must build ethical systems because it is **the right thing to do** — not because regulations demand it, not because a manager requires it, and not because a data breach would be bad publicity.

- A security engineer who writes robust encryption code because they believe in protecting users' right to privacy is acting from duty — this has moral worth.
- A security engineer who writes the same code only because their company will be fined for non-compliance is acting from self-interest — same technical output, but no moral worth.
- Whistleblowing — reporting unethical practices within one's organization — is a duty under Kantian ethics, even when it is personally costly.

**Conclusion:** Kant's Categorical Imperative gives CSE professionals a powerful ethical compass. It reminds them that some actions are simply wrong — not because they produce bad outcomes, but because they violate the rational dignity of human beings. In a field where technical capability often runs far ahead of ethical reflection, Kantian ethics provides the non-negotiable boundaries that protect humanity from technological exploitation.

---

## QUESTION 9

**"Discuss Aquinas's Natural Law Theory and its significance in the history of ethics. How does it remain relevant today?"**

---

### PART A: St. Thomas Aquinas and the Natural Law Theory

**Background — St. Thomas Aquinas (1225–1274):** St. Thomas Aquinas is the second and most philosophically sophisticated thinker of the Medieval Period of ethical thought (500–1500 AD). The Medieval Period was defined by the dominance of Christianity across Europe, where ethics was understood primarily as obedience to God's will as set forth in the Bible and interpreted by the Church.

Aquinas's greatest works were **Summa Theologica** and **Summa Contra Gentes** — comprehensive attempts to demonstrate the harmony between Aristotelian reason and Christian faith.

**The Historical Context of Aquinas's Achievement:** In the early Medieval Period, St. Augustine's Platonic-Christian system dominated ethical thought, and Aristotle had been largely forgotten in Western Europe. Aquinas **rediscovered Aristotle**, Christianized his philosophy, and made it the philosophical basis of the later Middle Ages. This was a pivotal intellectual achievement.

---

### The Natural Law Theory:

Aquinas's central contribution is his **Natural Law Theory** — the idea that God's eternal moral law is not only found in Scripture but is also **written in the heart of every human being** and can be discovered through reason and cultivated through conscience.

This means:

- Moral truth is **universally accessible** — not only to Christians or those who have read the Bible, but to all rational human beings
  - Reason is a valid and reliable path to moral knowledge
  - Conscience is the faculty through which natural law is recognized and applied in daily life
- 

### The Two Dimensions of Ethics (Aquinas):

Aquinas taught that ethics has **two dimensions** — the natural and the theological — which are complementary, not contradictory:

**Dimension 1 — Natural Ethics:** Natural ethics, as detailed by Aristotle, consists of the **development of reason and practice in living morally**. It leads to **earthly happiness**. This dimension is accessible to all people through reason alone — no special revelation or religious belief is required.

**Dimension 2 — Theological Ethics:** Theological ethics consists of achieving the virtues of **faith, hope, and charity** through God's grace. It leads to **eternal life with God**. This dimension goes beyond what reason alone can achieve.

**The Relationship Between the Two:** Aquinas demonstrated that natural ethics and theological ethics are not in conflict — they are **complementary levels of the same moral reality**. Reason and faith are in harmony. This was a revolutionary claim in an age when many believed that secular knowledge was dangerous to faith.

---

### The Significance of Aquinas's Natural Law Theory:

**1. Harmony of Reason and Faith:** Aquinas resolved the perceived conflict between reason and religious belief. He showed that rational inquiry into ethics and theological obedience lead to compatible conclusions. This intellectual reconciliation was the foundation of scholastic philosophy and laid the groundwork for universities and formal education in Europe.

**2. Ethics Accessible Through Reason:** By arguing that natural law can be discovered by reason and does not require religious revelation, Aquinas made ethics universally accessible. This has profound implications — it means that people of all faiths and cultures share a common moral foundation accessible through rational reflection.

**3. Path Toward Scientific Ethics:** Most significantly, by allowing people to turn to **secular knowledge without guilt**, Aquinas prepared the way for the emergence of a more scientific view of humanity and of ethics. He effectively gave permission for future generations to pursue philosophy, science, and rational inquiry as legitimate paths to truth — without abandoning faith. This intellectual opening contributed to the conditions that eventually produced the Scientific Revolution and the Modern Period of ethical thought.

---

## **PART B: Relevance of Aquinas's Natural Law Theory Today**

**1. Foundation of International Law and Human Rights:** The idea that there is a **universal moral law accessible to all rational beings** — regardless of their religion or culture — is the philosophical foundation of modern international human rights law. The Universal Declaration of Human Rights (1948) assumes that certain rights belong to all humans by virtue of their rational nature — a fundamentally Aquinian idea.

*Relevance to CSE:* International cybersecurity law, digital rights frameworks, and data protection regulations like GDPR are grounded in the idea that all people — regardless of nationality — have fundamental digital rights that no government or corporation may violate. This is Natural Law theory applied to the digital age.

**2. Reason and Ethics in Technology:** Aquinas's argument that ethics is accessible through reason — not only through religious or cultural authority — supports the development of **secular technology ethics frameworks** that can be applied universally across cultures.

*Relevance to CSE:* When global technology companies like Google, Microsoft, or Facebook develop AI ethics principles, they are implicitly applying a form of Natural Law reasoning — seeking universal ethical standards through rational analysis rather than religious or national authority.

**3. Conscience as Ethical Guide:** Aquinas's emphasis on conscience as the application of Natural Law to specific situations maps directly onto the modern concept of **professional conscience** — the internal moral guide that tells an engineer when something is wrong, even when no rule explicitly prohibits it.

*Relevance to CSE:* A CSE professional who refuses to build a surveillance system that would be used to oppress minorities — even when legally permitted and economically incentivized — is exercising conscience in exactly the way Aquinas described: recognizing the natural law written in their heart and acting accordingly.

**4. Complementarity of Technical and Ethical Knowledge:** Just as Aquinas argued that secular knowledge and theological knowledge are complementary, not contradictory, the same principle applies to technical knowledge and ethical knowledge in CSE. Engineering expertise and ethical reflection are not in conflict — they are two dimensions of the same professional responsibility.

**Conclusion:** Aquinas's Natural Law Theory, developed in the 13th century, continues to provide the philosophical foundation for universal human rights, international law, professional conscience, and secular ethics frameworks. For CSE professionals, its most enduring lesson is that moral knowledge is accessible to all rational beings — which means that every engineer, regardless of their background, has both the capacity and the responsibility to reason ethically about the technology they build.

---

# QUESTION 10 (NEW)

## "Discuss Hobbes's Social Contract Theory and its relevance to digital governance and cybersecurity."

---

### PART A: Hobbes's Social Contract Theory

**Background — Thomas Hobbes (1588–1679):** Thomas Hobbes belongs to the Modern Period of ethical thought. He was the **first thinker to systematically approach ethics from a scientific viewpoint**. His major philosophical work was **Leviathan** (1651). Hobbes applied the principle of **mechanistic materialism** — central to the physics of his time — to ethics, arguing that human behavior follows predictable, mechanical patterns just as physical objects do.

**The State of Nature:** Hobbes began with a thought experiment: what would human life be like without society or government — in its "natural" or primitive state?

His conclusion was stark: without civil society, human beings are driven entirely by **self-interest and self-preservation**. In this natural condition:

- There is no concept of right or wrong — morality does not yet exist
- Every person competes with every other for survival and resources
- Life would be, in his famous words: *"solitary, poor, nasty, brutish, and short"*

**The Social Contract:** To escape this condition of perpetual conflict, Hobbes argued that human beings collectively agree — through an implicit **Social Contract** — to give up some of their freedom and transfer power to a **Sovereign** (a ruler or governing body). In return, the Sovereign guarantees order, safety, and the conditions for social life.

Key elements:

- Morality and the concepts of right and wrong **only begin to exist** when civil society is formed — they are products of social agreement, not natural facts
- The Sovereign — whether a monarch or an assembly — must have sufficient authority to control individual self-interest and prevent conflict
- Hobbes believed in the **Golden Rule** (treat others as you wish to be treated) but argued that people cannot be trusted to follow it voluntarily — hence the necessity of sovereign authority

**Reaction to Hobbes:** Many ethicists disagreed with Hobbes's bleak view of human nature. They argued that human beings possess a **moral faculty** beyond mere self-interest — variously described as:

- **Intuition:** A natural sense of right and wrong
  - **Moral sense:** A natural affection for virtue
  - **Conscience:** An internal guide to ethical conduct
-



## PART B: Relevance to Digital Governance and Cybersecurity

**1. The Internet as a "State of Nature":** Hobbes's description of the natural state — where self-interest dominates and there are no enforced rules — is a remarkably accurate description of the early internet and continues to describe the "dark web" today. Without governance, digital spaces become environments of exploitation, fraud, harassment, and cybercrime.

*Application:* Just as Hobbes argued that sovereign authority is necessary to create social order from natural chaos, **digital governance frameworks** — national cybersecurity laws, international internet governance bodies, platform content moderation — are the modern equivalents of the Hobbesian Sovereign. They impose order on what would otherwise be digital anarchy.

**2. Cybersecurity as Social Contract:** Hobbes's Social Contract implies that citizens give up certain freedoms (absolute privacy, anonymous harmful behavior) in exchange for security. This trade-off is at the heart of modern cybersecurity debates:

- Governments argue for **backdoors in encryption** — some privacy surrendered for national security (Hobbesian)
- Civil libertarians argue this violates fundamental rights — a rejection of Hobbes in favor of moral faculty theories

*Example:* The ongoing debate about end-to-end encryption between governments and technology companies is fundamentally a Hobbesian argument: how much privacy (individual freedom) should citizens surrender to the Sovereign (government) for collective security?

**3. Platform Governance:** Social media platforms function as quasi-Hobbesian sovereigns — they establish rules (community standards), enforce them (content moderation), and penalize violations (banning users). Without this governance, platforms descend into the equivalent of Hobbes's state of nature: harassment, misinformation, and exploitation dominate.

*Bangladesh example:* The spread of communal violence-inciting content on Facebook in Bangladesh demonstrates the consequences of insufficient platform governance — a digital state of nature where self-interest (engagement, outrage) dominates over collective safety.

**4. The Moral Faculty Response in CSE:** The ethicists who reacted against Hobbes — arguing that humans have intuition, moral sense, and conscience — remind CSE professionals that governance frameworks alone are insufficient. Engineers must internalize ethical values, not merely comply with external rules.

**Conclusion:** Hobbes's Social Contract Theory provides a powerful framework for understanding why digital governance, cybersecurity regulation, and platform content moderation are ethically necessary — not merely politically convenient. At the same time, the reaction to Hobbes reminds us that external rules must be complemented by internal conscience for technology to truly serve humanity.

---

# QUESTION 11 (NEW)

**"Discuss the role of Hedonism and Duty-based Ethics in the history of ethical thought. Compare them and evaluate their relevance to CSE."**

---

## **PART A: Hedonism in Ethical History**

**Definition:** Hedonism is the ethical theory that **pleasure is the highest good** and the proper aim of human life. The measure of any action is the amount of pleasure it brings. The word comes from the Greek *hedone* meaning pleasure.

**Classical Hedonism — The Cyrenaics:** The Cyrenaics were an early Greek school who argued that any pleasure — physical or mental — is inherently good. They emphasized immediate, present pleasure as the goal of action.

**Refined Hedonism — The Epicureans:** Epicurus (341–270 BC) made an important refinement. He distinguished between:

- **Natural pleasures:** Peace of mind, absence of hunger, simple friendship — these are genuinely good
- **Unnatural pleasures:** Greed, lust, excessive ambition — these ultimately cause more pain than pleasure

Epicurus approved only natural pleasures. True happiness comes from tranquility (*ataraxia*) — freedom from anxiety — not from the pursuit of intense sensory pleasure.

**Modern Hedonism — David Hume and John Stuart Mill:** David Hume (1711–1766) incorporated hedonistic elements — arguing that pleasure and utility are the standards of moral judgment. Mill's Utilitarianism is explicitly hedonistic: "pleasure or happiness is the standard of moral judgment," and the goal is "the greatest happiness for the greatest number."

---

## **PART B: Duty-Based Ethics (Deontology) in Ethical History**

**Classical Duty Ethics — The Stoics:** The Stoics (ancient Greece and Rome) stood in direct opposition to the Hedonists. They argued that the path to the good life is not pleasure but **virtue through rational self-discipline**. Emotions and desires must be overcome; duty as revealed by reason must be followed.

**Modern Duty Ethics — Immanuel Kant:** Kant (1724–1804) developed the most sophisticated duty-based ethical theory. His Deontology argues that:

- Morality is grounded in **pure reason**, not emotion or pleasure
- **Good Will** — acting from duty — is the only unconditionally good thing
- The **Categorical Imperative** provides universal moral commands that apply regardless of consequences
- Humans are rational beings whose dignity demands they never be treated merely as means

---

## PART C: Comparison

| Aspect                      | Hedonism                                           | Duty-Based Ethics (Deontology)                     |
|-----------------------------|----------------------------------------------------|----------------------------------------------------|
| Foundation of morality      | Pleasure and happiness                             | Reason and duty                                    |
| What makes an action right? | It produces pleasure/good consequences             | It is performed from duty/good will                |
| Role of emotion             | Central — pleasure is the guide                    | Must be excluded — reason commands                 |
| On lying                    | Acceptable if it produces greater pleasure         | Always wrong — violates universal law              |
| Key thinkers                | Cyrenaics, Epicureans, Hume, Mill                  | Stoics, Kant                                       |
| Strength                    | Practical, measurable, human-centered              | Principled, consistent, protects dignity           |
| Weakness                    | Can justify harmful acts if they maximize pleasure | Can be rigid and impractical in complex situations |

---

## PART D: Relevance to CSE

**Hedonism (Utilitarian form) in CSE:** Utilitarian/hedonistic thinking is used when engineers must make trade-off decisions — maximizing overall benefit while minimizing total harm.

- Open-source software: Releasing code publicly causes some risk (bad actors can study it) but produces vastly more pleasure/benefit (global community can improve and use it) — Utilitarian decision
- AI healthcare tools: A medical AI that is 98% accurate might cause harm to 2% of users — a Utilitarian analysis weighs the massive benefit to the 98% against the harm to the 2%
- Public data sharing for research: Some privacy loss produces large gains in scientific knowledge — a hedonistic calculation

*Risk of pure Hedonism in CSE:* It can justify exploitative practices. Cambridge Analytica's data harvesting "maximized utility" for political clients — while harming millions of users. This shows why pure Hedonism without constraints is dangerous.

**Duty-Based Ethics in CSE:** Kantian/Stoic duty-ethics provides the absolute rules that protect against the excesses of pure outcome-thinking.

- User data must never be sold without consent — regardless of how much revenue it would generate
- Security vulnerabilities must be disclosed responsibly — it is a professional duty
- Discriminatory AI must be redesigned — not deployed even if it is technically impressive or economically profitable

**Synthesis:** The most mature CSE ethics combines both: **Hedonism/Utilitarianism provides the framework for navigating trade-offs, while Duty Ethics provides the non-negotiable boundaries.** Kant draws the red lines; Mill helps navigate the grey areas within them.

**Conclusion:** The ancient debate between pleasure-based and duty-based ethics — begun in the Classical Period between Epicureans and Stoics — continues in modern CSE ethics. Understanding both traditions equips engineers to make nuanced decisions that are both practically effective and morally principled.

---



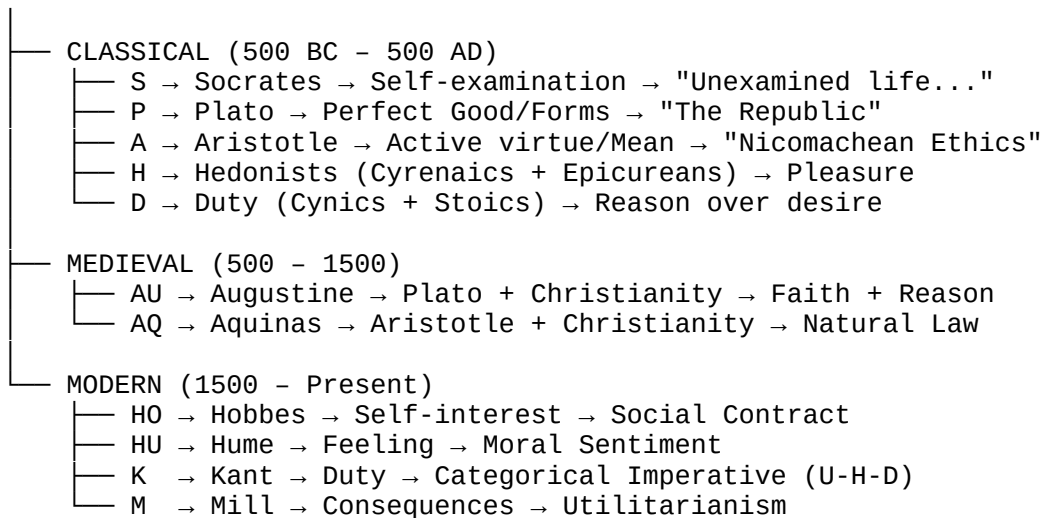
## UPDATED MEMORIZATION GUIDE

*(Combines all questions including newly added ones)*

---

### MASTER MEMORY MAP

#### ETHICS HISTORY



### ONE-CARD REVISION SHEET

**ETHICS:** Greek *Ethos* → systematic study of right/wrong → forms good people → art of human living

**SOCRATES:** Reason + Self-knowledge | "Unexamined life..." | Socratic Method | No writings

**PLATO:** The Good + Forms | The Republic | Philosopher-King | Denied bodily pleasure

**ARISTOTLE:** Virtue = Mean | Daily life ethics | Nicomachean Ethics | "No form without matter"

**AUGUSTINE:** Plato + Christianity | Earthly vs Eternal good | Faith needed for virtue | Confessions + City of God

**AQUINAS:** Aristotle + Christianity | Natural Law | Reason + Faith in harmony | Summa Theologica | Path to scientific ethics

**HOBBS:** Self-interest | State of nature = chaos | Social Contract → Sovereign | Leviathan

**HUME:** Feeling over reason | Two-sided judgment (objective + subjective) | Moral sentiment

**KANT:** Duty (Deontology) | Good Will | Categorical Imperative: Universal Law + Humanity as End + Duty gives worth

**MILL:** Consequences (Utilitarianism/Teleology) | Greatest happiness | Hedonist | External + Internal motivation

**CSE ISSUES (DC-SAI-WE):** D=Data Privacy | C=Cybersecurity | S=Social Media | A=AI Fairness | I=Intellectual Property | W=Workplace | E=Environment

**BANGLADESH ISSUES:** Digital divide | 5 crore data breach | Social media misuse | Automation job loss

# Exam-Ready English Examples for DCSEAIEW

---

## D — Data Privacy & Surveillance

### 2.1.1 Unauthorized Data Collection

In 2023, the personal data of over 50 million Bangladeshi citizens — including NID numbers, phone numbers, and addresses — was found exposed online through a government website vulnerability. Citizens had no idea their data was being collected or stored insecurely.

### 2.1.2 Surveillance Systems

China's Social Credit System monitors citizens through facial recognition cameras, tracking their behavior in public spaces. People with low scores face travel bans and restricted access to services — a chilling example of state surveillance overriding individual privacy.

### 2.1.3 Misuse of Personal Information

In the Cambridge Analytica scandal (2018), personal data from 87 million Facebook users was harvested without consent and used to psychologically profile voters during the US Presidential Election, manipulating political decisions at a massive scale.

### 2.1.4 Data Breaches Due to Poor Security

In 2017, Equifax — one of the largest credit reporting agencies — suffered a data breach exposing the financial records of 147 million Americans due to an unpatched software vulnerability. Engineers had ignored the security warning for months.

---

## C — Cybersecurity & Responsible Security Practices

### 2.2.1 Ethical Problems in Penetration Testing

A cybersecurity firm hired to test a hospital's network discovered patient records were accessible. Without proper authorization documentation, sharing this finding could violate confidentiality laws. This illustrates why **prior consent and defined scope** are essential in ethical pen testing.

### 2.2.2 Handling Zero-Day Vulnerabilities

In 2021, the Microsoft Exchange Server zero-day vulnerability was exploited by hackers before Microsoft could release a patch. Over 250,000 servers worldwide were compromised. Responsible disclosure and **proactive multilayer prevention** could have reduced the damage.

### 2.2.3 Ethical vs Malicious Hacking

Kevin Mitnick, once the world's most wanted hacker (Black Hat), later became a cybersecurity consultant (White Hat). The difference? **Authorization and intent.**

Ethical hackers operate within legal boundaries with written permission; malicious hackers do not.

#### 2.2.4 Engineers Discovering Security Flaws

A software engineer at Yahoo discovered in 2014 that 3 billion user accounts were compromised. However, Yahoo delayed public disclosure for **two years**, violating the ethical duty of prompt response and putting users at ongoing risk.

---

## S — Social Media Ethics

### 2.3.1 Spread of Harmful Content

In 2017, Facebook's algorithm amplified anti-Rohingya hate speech in Myanmar, contributing to real-world ethnic violence. UN investigators stated that Facebook played a "determining role" in spreading content that incited genocide. The platform failed to moderate harmful content in a minority language.

### 2.3.2 Manipulation Through Algorithms

Netflix's recommendation algorithm and YouTube's autoplay feature are designed to maximize watch time — not user wellbeing. Studies show YouTube's algorithm frequently leads users from moderate content toward increasingly extreme videos, radicalizing opinions without users realizing it.

### 2.3.3 Echo Chambers and Polarization

During the 2022 Brazilian election, WhatsApp groups circulated massive volumes of misinformation. Users who received false content only interacted with others who believed the same things — a classic echo chamber — making it nearly impossible to correct misinformation once it spread.

### 2.3.4 Data Harvesting for Targeted Advertising

Facebook tracks users even when they are **not on Facebook** — through invisible "Like" buttons embedded on third-party websites. This harvested behavioral data is sold to advertisers for micro-targeted ads, often exploiting emotional vulnerabilities without users' genuine informed consent.

---

## A — AI & Algorithmic Fairness

### 2.4.1 Bias in Machine Learning Datasets

Amazon built an AI recruitment tool trained on 10 years of resumes — mostly from men. The system learned to **penalize resumes containing the word "women's"** (e.g., "women's chess club"). Amazon scrapped the tool in 2018 after discovering it systematically discriminated against female applicants.

### 2.4.2 Discrimination in Automated Decision Systems

The COMPAS algorithm used in US courts to predict criminal recidivism was found to be **twice as likely to falsely flag Black defendants as future criminals** compared to white defendants. Judges relied on this score in sentencing decisions, embedding racial bias into the justice system.

### 2.4.3 Lack of Transparency — Black Box System

When a person is denied a bank loan by an AI system, they often receive no explanation. The AI processes hundreds of variables and produces a decision — but neither the applicant nor even the bank's own staff can explain exactly why the loan was denied. This lack of transparency violates principles of fairness and accountability.

### 2.4.4 Job Displacement Due to Automation

McKinsey Global Institute estimates that by 2030, automation could displace up to **375 million workers** globally. In Bangladesh's garment industry, automated sewing machines already threaten millions of jobs held mostly by women — raising serious questions about who bears the cost of technological progress.

### 2.4.5 Deepfakes and Misinformation

In 2023, AI-generated deepfake audio clips of political leaders making inflammatory statements were circulated on social media in several countries before elections. These were entirely fabricated but were convincing enough to spark public outrage, demonstrating how AI can be weaponized for misinformation at scale.

---

## I — Intellectual Property & Software Piracy

### 2.5.1 Ownership of Code

When a developer writes code as part of their employment contract, the **company typically owns that code** — not the developer. This became a major legal dispute when a former Google engineer took proprietary self-driving car code to Uber. Uber paid \$245 million in settlement, highlighting the legal weight of code ownership.

### 2.5.2 Open-Source vs Proprietary Conflicts

Oracle sued Google for using Java APIs in Android without a license, claiming \$9 billion in damages. After a decade-long legal battle, the US Supreme Court ruled in Google's favor in 2021 — a landmark case defining the boundaries between open-source principles and proprietary IP rights.

### 2.5.3 Plagiarism in Academic or Professional Coding

A computer science student submitted an assignment using code copied from GitHub without attribution. The professor detected it using plagiarism detection software. In professional settings, copying proprietary code without a license can lead to lawsuits, job termination, and reputational damage.

### 2.5.4 Patents and Licensing Issues

Apple and Samsung engaged in one of the most expensive patent wars in tech history (2011–2018), with Apple claiming Samsung copied iPhone's rounded-rectangle design



and touch interface. The case resulted in over **\$1 billion** in damages being awarded to Apple, illustrating how seriously IP law treats design and software patents.

---

## E — Environmental & Sustainability Concerns

### 2.6.1 High Energy Consumption of Data Centers and Blockchain

Bitcoin mining alone consumes approximately **150 terawatt-hours of electricity per year** — more than the entire country of Argentina. Google's data centers consume enough electricity to power a mid-sized city. The carbon footprint of the internet and digital infrastructure is now larger than the aviation industry.

### 2.6.2 E-Waste from Rapid Hardware Obsolescence

Agbogbloshie in Accra, Ghana has become one of the world's largest e-waste dumpsites. Old electronics from Europe and North America — computers, phones, and cables — are shipped there illegally. Local workers, including children, burn the waste to extract copper, suffering severe lead and mercury poisoning, causing cancer and neurological damage.

### 2.6.3 Sustainable Software Practices

"Green coding" is an emerging practice where developers optimize software to use **less processing power**, reducing energy consumption. For example, inefficient code that runs unnecessarily complex operations forces servers to consume more electricity. Microsoft and Google have both committed to carbon-neutral cloud computing by 2030 as part of sustainable software responsibility.

---

## W — Workplace & Professional Ethics

### 2.7.1 Conflicts of Interest

A software engineer working for a government e-governance platform discovers that their company is also privately contracted by a competing political party to build voter data tools. Using the same data access for both creates a serious **conflict of interest** that compromises public trust and professional integrity.

### 2.7.2 Whistleblowing Dilemmas

Frances Haugen, a former Facebook product manager, leaked thousands of internal documents in 2021 proving that Facebook knew Instagram caused mental health harm to teenage girls — but chose profit over safety. She testified before the US Senate. Her case shows the enormous personal and professional **risk of whistleblowing**, but also its necessity for public good.

### 2.7.3 Ethical Responsibilities During Product Development

Engineers at Boeing were reportedly aware of safety concerns with the 737 MAX autopilot system but faced pressure to meet deadlines. The plane crashed twice, killing **346 people**. This is a defining case of how ignoring ethical responsibility during product development leads to catastrophic real-world consequences.

#### 2.7.4 Handling Pressure from Employers

Volkswagen engineers were aware that their vehicles were programmed to cheat emissions tests — performing better during testing than in real-world driving. Instead of reporting this, engineers complied under management pressure. The scandal eventually exposed **11 million fraudulent vehicles** globally and cost Volkswagen over \$30 billion in fines and settlements.